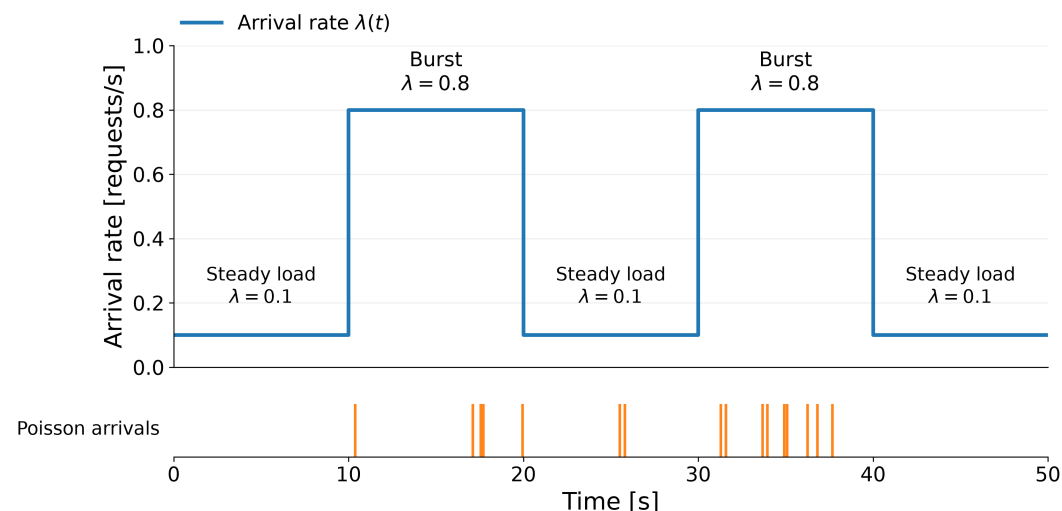


# Adaptive Scheduling Makes Qwen3.8-27B NVFP4 Faster

The latency-optimal vLLM scheduling budget changes with serving pressure. P-PAS adapts it dynamically.



## Why adapt the scheduling budget?

Controlled deterministic workloads show that the best fixed budget changes with serving pressure.

| Load      | MBT 2048      | MBT 768       | Winner      |
|-----------|---------------|---------------|-------------|
| 0.1 req/s | <b>2.98 s</b> | 3.19 s        | <b>2048</b> |
| 0.8 req/s | 11.89 s       | <b>9.81 s</b> | <b>768</b>  |

## P-PAS on the dynamic workload

Five Poisson traces with identical arrival patterns per scheduler (5 seeds, alternating 0.1  $\leftrightarrow$  0.8 req/s).

| Scheduler               | Avg. latency   | vs. P-PAS           |
|-------------------------|----------------|---------------------|
| vLLM default (MBT 2048) | 13.73 s        | <b>P-PAS -14.7%</b> |
| Fixed MBT 768           | 12.19 s        | <b>P-PAS -3.9%</b>  |
| <b>P-PAS</b>            | <b>11.71 s</b> | <b>best</b>         |

**14.7% lower end-to-end latency vs. vLLM default**

vLLM 0.27.1 • P-PAS • Qwen3.8-27B NVFP4 • RTX 5090